WEBBY

AI-Powered Adaptive Accessibility for the Web

TEAM :- **TRIDENT**

> “Don't adapt yourself to the website. Let the website adapt to you.”

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PROBLEM STATEMENT

Web accessibility is still inconsistent and highly context-dependent.

Users experience different accessibility barriers across websites because each site has different layout, typography, contrast, spacing, motion, and interaction patterns. Even when accessibility preferences are known, users may still need to repeatedly adjust websites to reach a comfortable experience.



Evidence

95.9%
of the top one million home pages had detectable WCAG 2 failures.

Source: WebAIM Million 2026

83.9%
of home pages contained low-contrast text.

Source: WebAIM Million 2026

1,437
average elements per home page, showing increasing webpage complexity.

Source: WebAIM Million 2026

The Core Problem

> **Accessibility settings exist, but they are often too coarse to reflect both the individual user and the context of each webpage.**

PROBLEM IDENTIFICATION

The Accessibility Gap

1 Different Websites, Different Barriers

Websites vary in layout, typography, contrast, spacing, navigation, and interaction patterns, creating different accessibility challenges.

2 Repetitive Manual Adjustments

Users may repeatedly change text size, spacing, contrast, motion, focus, or layout preferences.

3 Accessibility Needs Are Individual

Different users require different combinations of accessibility preferences rather than a single fixed mode.

4 Context Matters

The same accessibility preference may need to be applied differently depending on the webpage.

WEBBY's Positioning



User-controlled



Persistent



Context-aware

CORE GAP

The missing layer is one that understands both the user and the webpage before deciding how accessibility preferences should be applied.

PROPOSED SOLUTION

WEBBY is a proposed browser-based adaptive accessibility layer that combines rule-based analysis, AI-powered semantic understanding, and a persistent Personal Accessibility Profile.

01

Understand the Webpage

Rule-based checks examine text size, spacing, contrast, motion, and focus styling.

02

Understand the Context

AI helps interpret main content, clutter, unlabeled controls, images, and dense information.

03

Understand the User

WEBBY retrieves the user's Personal Accessibility Profile and learned Accessibility Twin.

04

Decide the Adaptation

Webpage context and user preferences are combined to determine appropriate changes.

05

Learn From Explicit Feedback

User adjustments update the Twin so future starting configurations can become more personalized.

CORE FLOW

Webpage → Privacy Filter → Rules + AI Analysis → User Profile / Twin → Adaptation Decision → Validation → Adapted Webpage

KEY IDEA

WEBBY transforms accessibility from generic settings into a personalized, context-aware web experience.

KEY FEATURES

1



AI-Powered Web Understanding

Uses semantic understanding to interpret elements that cannot be handled reliably through simple style checks alone.

2



Granular Personal Accessibility Profile

Stores user-selected preferences such as text scale, line spacing, contrast preference, motion sensitivity, focus assistance, and interaction preferences.

3



Context-Aware Adaptation

The same preference is applied differently depending on the webpage.

Example: a strong contrast preference may darken text on a light webpage, while an already-dark webpage may need a different adjustment instead of simply inverting colours.

4



Learn From Me

Explicit user adjustments become preference signals that improve future starting configurations.

5



Accessibility Twin

The Twin combines the user's Profile, learned adjustments, per-site overrides, and patterns across different page contexts.

EXPECTED IMPACT

Reduced Manual Effort

Users may need fewer repeated adjustments when moving between websites.

Greater Personalization

Accessibility can be based on individual preference combinations rather than broad fixed settings.

More Consistent Web Experience

A persistent profile can provide a better starting point across different websites.

Better Context Awareness

Adaptations can account for the actual structure and characteristics of the current webpage.

PROPOSED EVALUATION METRICS

- Manual adjustments required on a new website
- Time taken to reach comfortable settings
- Contrast and spacing checks before vs. after adaptation
- User feedback from people with visual or reading accessibility needs

LONG-TERM VISION

Make accessibility a persistent, personalized layer of the user's web experience.

PROPOSED TECHNOLOGY STACK

1 Browser Extension

Runs on the user's browser and adapts webpages based on the user's preferences.

Chrome / Chromium | Manifest V3 (Chrome Extension) | TypeScript | Content Scripts

- Manifest V3 (Chrome / Chromium)
- Content scripts to capture and analyze webpages
- DOM & CSS manipulation for real-time adaptation
- MutationObserver to handle dynamic content
- Use extension local storage for quick access to settings

2 Web Adaptation Engine

Analyzes and modifies webpage elements using rules and validated instructions.

DOM Analysis | CSS / DOM Manipulation | Mutation Observer | Rule-Based Checks

- Detect and analyze text size, contrast, spacing, motion, and focus styles
- Apply accessible CSS transformations
- Handle dynamic content changes
- Validate all adaptations before applying
- Ensure changes maintain website functionality

3 AI Engine

Provides semantic understanding of webpage content and helps with context-aware decisions.

Node.js | Express | LLM API (OpenAI)

- Node.js / Express backend
- LLM API for semantic and contextual analysis
- Identify main content vs. clutter
- Interpret ambiguous or unlabeled elements (including images and controls)
- Support context-aware adaptation decisions

4 WEBBY Dashboard

Allows users to manage their profile, view their Accessibility Twin, and track preference history.

Next.js | Twin View | Profile Management

- Next.js frontend
- Visual Accessibility Twin view
- Manage and customize accessibility profile
- View preference history and per-site overrides

5 Data Layer

Stores user preferences and optional sync data.

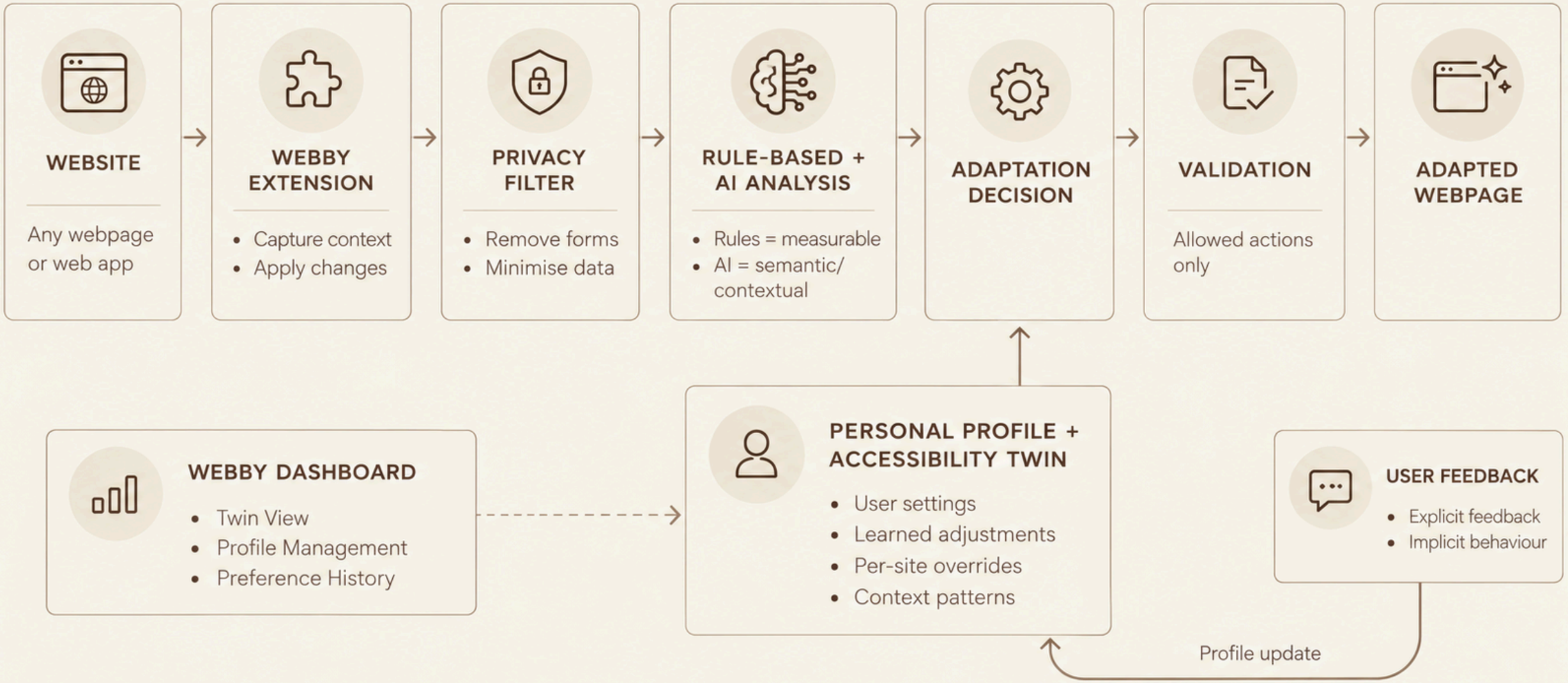
Local Storage (by default) | Supabase (optional sync)

- Store profile locally on the user's device
- Optional profile synchronization via Supabase
- User authentication
- Accessibility preferences and learning data
- Sync only with user consent

CORE PRINCIPLE

Rules handle measurable adaptations immediately; AI handles semantic and contextual decisions.

PROPOSED IMPLEMENTATION APPROACH



PRIVACY & SAFETY

User Profile

Accessibility preferences can be sensitive because they may reveal personal accessibility needs.

On-device by default; sync only with user consent.

Minimal AI Input

Only the information required for accessibility analysis should be shared with the AI engine.

Sensitive Content Protection

- Exclude passwords and form fields
- Avoid unnecessary page content
- Skip or restrict processing on sensitive pages
- AI-assisted processing is transparent and opt-in.

AI Safety

The AI returns validated adaptation instructions, not executable code.

WEBBY applies only actions from a predefined and permitted set of accessibility transformations.

DESIGN PRINCIPLE

Minimize shared data, keep the user in control, and validate every AI-generated adaptation.

CHALLENGES & FUTURE SCOPE

KEY CHALLENGES

AI Latency & Cost

Semantic AI analysis may introduce delay and API cost.

Mitigation: immediate rule-based adaptations, compressed webpage summaries, and cached AI decisions.

Website Compatibility

DOM/CSS modifications may behave differently across complex websites.

Mitigation: reversible changes, validation, and website-specific overrides.

Assistive Technology Compatibility

Visual or structural changes must not interfere with screen readers, keyboard navigation, or semantic HTML.

Mitigation: preserve accessibility semantics and treat assistive-technology compatibility as a core design constraint.

Privacy & Security

Webpage content and accessibility preferences may contain sensitive information.

Mitigation: local-first profile storage, privacy filtering, opt-in synchronization, and validated AI outputs.

FUTURE SCOPE

- Broader support for screen-reader users
- Hearing and multimodal accessibility
- Mobile accessibility support
- More on-device AI processing
- Language-aware accessibility
- Cognitive-load and focus modes
- Expanded personalization across devices

FUTURE VISION

Evolve WEBBY from a browser accessibility layer into a persistent accessibility companion across the digital experience.